

Arnav Rawat

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About Me

Computer Science and Mathematics graduate passionate about building scalable software and ML systems. Experienced in full-stack development, algorithmic optimization, and real-time applications. Seeking opportunities to apply rigorous mathematical and programming skills to solve impactful engineering challenges.

Education

University of Illinois Urbana-Champaign

BS Math and Computer Science with High Distinction | GPA: 3.5

2021 – 2025

Experience

Independent Study – Mechanistic Interpretability

June 2025 – Present

- Learning techniques including Activation Patching, Linear Probes and Sparse Autoencoders
- Carried out experiments on local models including ablation and token prefill
- Hope to contribute towards advanced AI that is safe and beneficial for humanity

KDE Contributor

2022 – Present

- Work with developers on code development, reviews for Open Source Linux applications

Machine Learning Intern

June 2024 – August 2024

HGS Digital

Warrenville, IL

- Quickly adapted to new technologies to deliver AI solutions – including graph-based retrieval-augmented generation (RAG) systems that reduced hallucination rates by 8%
- Applied strong problem-solving skills to design and refine fraud detection models using ML outlier detection methods such as One-Class SVM, reducing false positives by 10% compared to baseline
- Partnered with teammates to troubleshoot and integrate solutions with Azure SQL, OpenAI APIs, and Azure Machine Learning Studio, ensuring reliable deployment

Digital Transformation Intern

June 2022 – July 2022

Pregis Inc

Bannockburn, IL

- Collaborated with the HR department to support a digital transformation by digitizing and making personnel documents fully searchable with OCR
- Applied statistical methods to validate recognized text for accuracy and completeness, such that 99% of text was correctly scanned

Projects

Agent With Retrieval: OpenAI, Vector Databases, Structured Outputs, React

- Built for retrieval across a large input corpus, through embeddings, clustering and linear operations in the latent space
- Interaction through a React interface, utilizing structured outputs for multiple use cases, including storytelling and programming
- Exploratory work on fine-tuning for optimizing subjective criteria

Uniform Job Scheduler: Formal specification, Docker, Operational Research

- Combinatorial optimizer (Google OR-Tools) used to approximate best schedule for scheduling bag-producing machines
- Improved bag throughput by ~5% in simulation

Skills

- **AI/ML**: Agentic AI, RAG, Transformers, CNNs, SVMs, clustering, OpenAI API
- **Languages**: C/C++, C#, Java, Python, Rust
- **Frameworks & Tools**: PyTorch, Flask, ASP.NET, React, Qt
- **Databases**: Azure SQL, MS SQL server, MySQL, MongoDB
- **Cloud Platforms**: Azure
- **CI/CD**: Github Actions, Gitlab pipelines, Unit Testing and Integration Testing